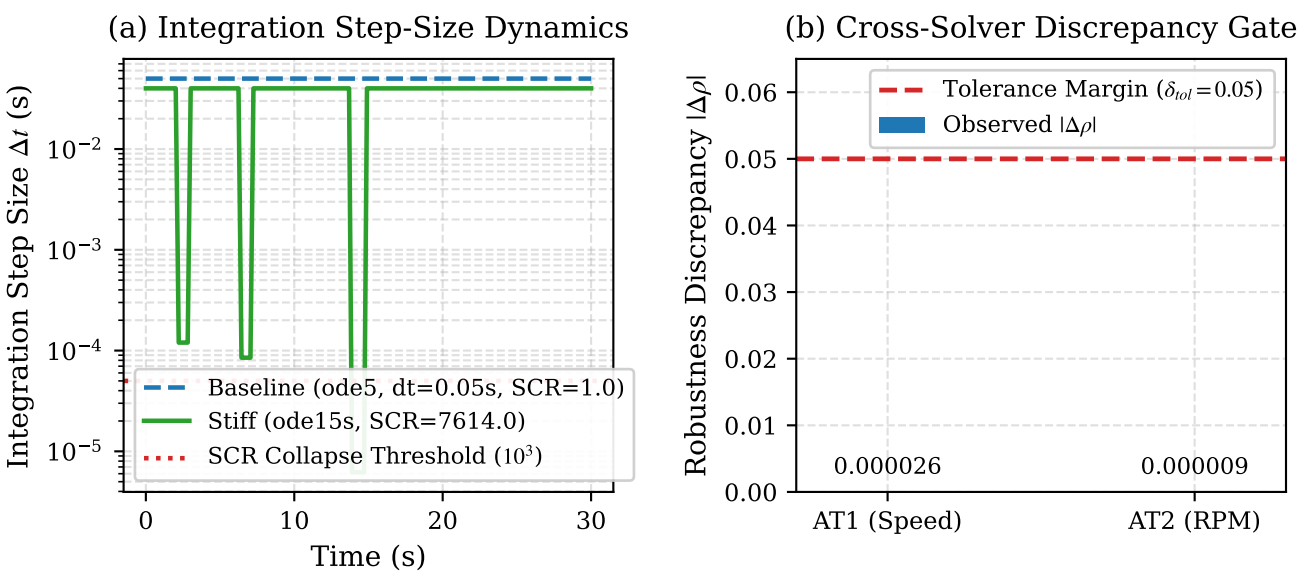
(a) Integration Step-Size Dynamics
Integration Step Size Δt (s)
Time (s)
Baseline (ode5, dt=0.05s, SCR=1.0)
Stiff (ode15s, SCR=7614.0)
SCR Collapse Threshold (10^3)
(b) Cross-Solver Discrepancy Gate
Robustness Discrepancy |Δρ|
Tolerance Margin (δ_tol = 0.05)
Observed |Δρ|
0.000026
0.000009
AT1 (Speed)
AT2 (RPM)